Task 1 Report: Sequence Analysis and Homology Search of Human Hemoglobin Subunit Beta (HBB)`

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Target Query ID: NP_000509.1 (NCBI Reference Sequence)

BLAST Request ID (RID): E6M2XSDZ014

1. Introduction & Background

Hemoglobin subunit beta (HBB) is a critical globular protein involved in oxygen transport from the respiratory organs to peripheral tissues across vertebrates. In humans, mutations in the HBB gene are associated with genetic disorders such as Sickle Cell Anemia and Beta-Thalassemia. Performing Sequence Analysis and BLAST (Basic Local Alignment Search Tool) enables the identification of sequence conservation, evolutionary relationships, and functionally critical residues across related mammalian species.

2. Methodology & Sequence Retrieval

- **Database Used:** NCBI (National Center for Biotechnology Information) Protein Database.
- **Protein Length:** 147 amino acids.
- **Algorithm:** blastp (protein-protein BLAST).
- **Matrix:** BLOSUM62.
- **Database:** ClusteredNR.

Table 1: Top BLAST Match Hits for Human HBB (NP_000509.1)

Target Description	Species / Organism	Max Score	Total Score	Query Cover	E-value	Percent Identity	Accession Number
Hb Monza protein	*Homo sapiens*	288	288	100 %	2×10^{-97}	86.47 %	XEF57200.1
beta-globin	*Homo sapiens*	277	277	100 %	2×10^{-93}	93.20 %	AAA35597.1
hemoglobin subunit beta-like	*Canis lupus familiaris* (Dog)	275	275	100 %	1×10^{-92}	89.80 %	NP_001257812.1
hemoglobin subunit beta-like	*Ailuropoda melanoleuca* (Panda)	273	273	100 %	8×10^{-92}	89.80 %	NP_001291812.1
hemoglobin subunit beta	*Rhynchonycteris naso* (Bat)	271	271	100 %	2×10^{-91}	89.12 %	KAM8815943.1
hemoglobin beta	*Tarsius sp.* (Primate)	259	259	99%	1×10^{-86}	84.93 %	765952B

4. Key Scientific Observations & Discussion

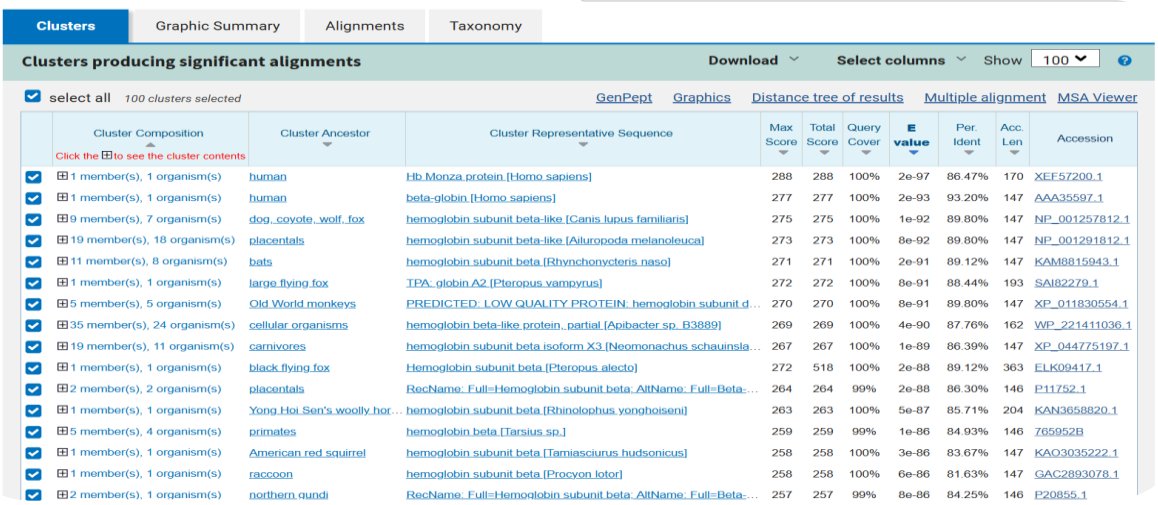
1. **Statistical Significance (E-Value):** All top alignments demonstrated extremely low Expect values ranging from 2×10^{-97} to 1×10^{-86}. An E-value near zero confirms that these matches are due to true biological homology rather than random chance.
2. **Query Coverage:** Every major hit showed **99% – 100% Query Coverage**, confirming that the overall secondary and tertiary fold architecture of HBB is preserved throughout mammalian species.
3. **Sequence Identity:** The percent identity ranged from **84.93% to 93.20%** across mammalian orders (Primates, Carnivora, Chiroptera), demonstrating significant evolutionary constraint on the amino acid sequence to maintain heme binding and oxygen release capabilities.

5. BLAST Results & Quantitative Metrics

The BLAST search returned significant alignments across diverse mammalian species, showing high evolutionary conservation. Below is the summary of key alignments extracted from RID E6M2XSDZ014.

6. Conclusion

The BLAST analysis of Human Hemoglobin Subunit Beta (NP_000509.1) successfully identified orthologous sequences across multiple species. The high sequence conservation and near-zero E-values emphasize the essential physiological role of HBB in mammalian survival.



Clusters | Graphic Summary | Alignments | Taxonomy

Clusters producing significant alignments — Download | Select columns | Show 100

select all *100 clusters selected* — GenPept | Graphics | Distance tree of results | Multiple alignment | MSA Viewer

Cluster Composition (Click the ⊞ to see the cluster contents)	Cluster Ancestor	Cluster Representative Sequence	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
⊞1 member(s), 1 organism(s)	human	Hb Monza protein [Homo sapiens]	288	288	100%	2e-97	86.47%	170	XEF57200.1
⊞1 member(s), 1 organism(s)	human	beta-globin [Homo sapiens]	277	277	100%	2e-93	93.20%	147	AAA35597.1
⊞9 member(s), 7 organism(s)	dog, coyote, wolf, fox	hemoglobin subunit beta-like [Canis lupus familiaris]	275	275	100%	1e-92	89.80%	147	NP_001257812.1
⊞19 member(s), 18 organism(s)	placentals	hemoglobin subunit beta-like [Ailuropoda melanoleuca]	273	273	100%	8e-92	89.80%	147	NP_001291812.1
⊞11 member(s), 8 organism(s)	bats	hemoglobin subunit beta [Rhynchonycteris naso]	271	271	100%	2e-91	89.12%	147	KAM8815943.1
⊞1 member(s), 1 organism(s)	large flying fox	TPA: globin A2 [Pteropus vampyrus]	272	272	100%	8e-91	88.44%	193	SAI82279.1
⊞5 member(s), 5 organism(s)	Old World monkeys	PREDICTED: LOW QUALITY PROTEIN: hemoglobin subunit d...	270	270	100%	8e-91	89.80%	147	XP_011830554.1
⊞35 member(s), 24 organism(s)	cellular organisms	hemoglobin beta-like protein, partial [Apibacter sp. B3889]	269	269	100%	4e-90	87.76%	162	WP_221411036.1
⊞19 member(s), 11 organism(s)	carnivores	hemoglobin subunit beta isoform X3 [Neomonachus schauinsla...	267	267	100%	1e-89	86.39%	147	XP_044775197.1
⊞1 member(s), 1 organism(s)	black flying fox	Hemoglobin subunit beta [Pteropus alecto]	272	518	100%	2e-88	89.12%	363	ELK09417.1
⊞2 member(s), 2 organism(s)	placentals	RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-...	264	264	99%	2e-88	86.30%	146	P11752.1
⊞1 member(s), 1 organism(s)	Yong Hoi Sen's woolly hor...	hemoglobin subunit beta [Rhinolophus yonghoiseni]	263	263	100%	5e-87	85.71%	204	KAN3658820.1
⊞5 member(s), 4 organism(s)	primates	hemoglobin beta [Tarsius sp.]	259	259	99%	1e-86	84.93%	146	765952B
⊞1 member(s), 1 organism(s)	American red squirrel	hemoglobin subunit beta [Tamiasciurus hudsonicus]	258	258	100%	3e-86	83.67%	147	KAO3035222.1
⊞1 member(s), 1 organism(s)	raccoon	hemoglobin subunit beta [Procyon lotor]	258	258	100%	6e-86	81.63%	147	GAC2893078.1
⊞2 member(s), 1 organism(s)	northern quoll	RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-...	257	257	99%	8e-86	84.25%	146	P20855.1